



EUROPEAN DDI USER CONFERENCE



METADATA MANAGEMENT

Using DDI and Colectica

EDDI 2015, Copenhagen

Agenda

09:00	Welcome
09:15	Introduction to Metadata, the DDI Information Model, and Colectica
09:45	Hands-on: Document a dataset
10:30	Break
11:00	Hands-on: Publish Metadata
11:30	Hands-on: Study descriptions
12:30	Lunch

13:30	Documenting Surveys and Data Collection
14:00	Hands-on: Questions and surveys
15:00	Break
15:30	Documenting longitudinal data
16:30	Recap
17:00	The End
18:00	Informal Get-Together at TapHouse

Introductions

- Colectica – a Brief History
 - ▣ Jeremy Iverson
 - Wisconsin Longitudinal Study
 - ▣ Dan Smith
 - IBM, University of Minnesota

First: A Quick Survey



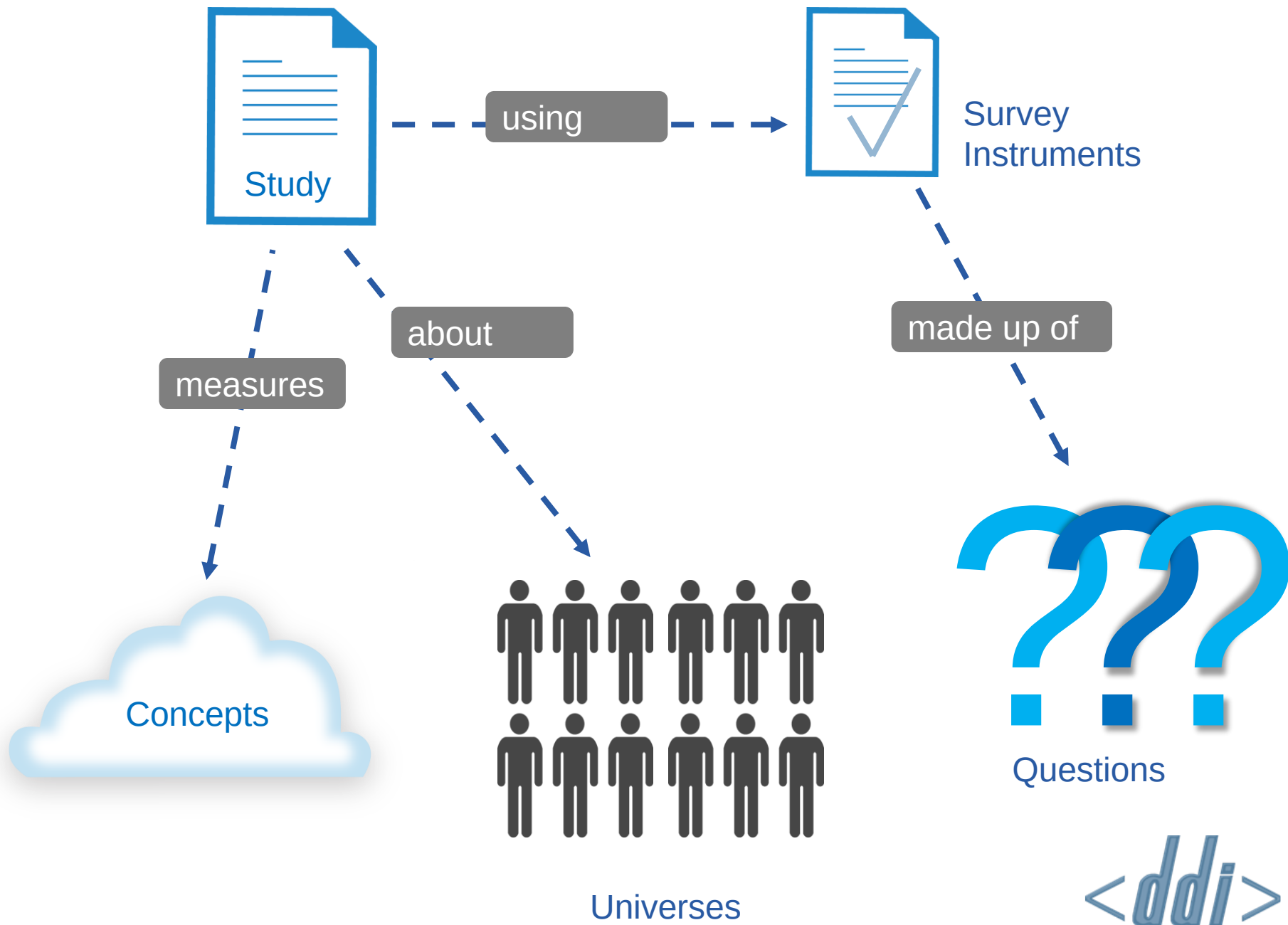
bit.ly/eddi2015-workshop

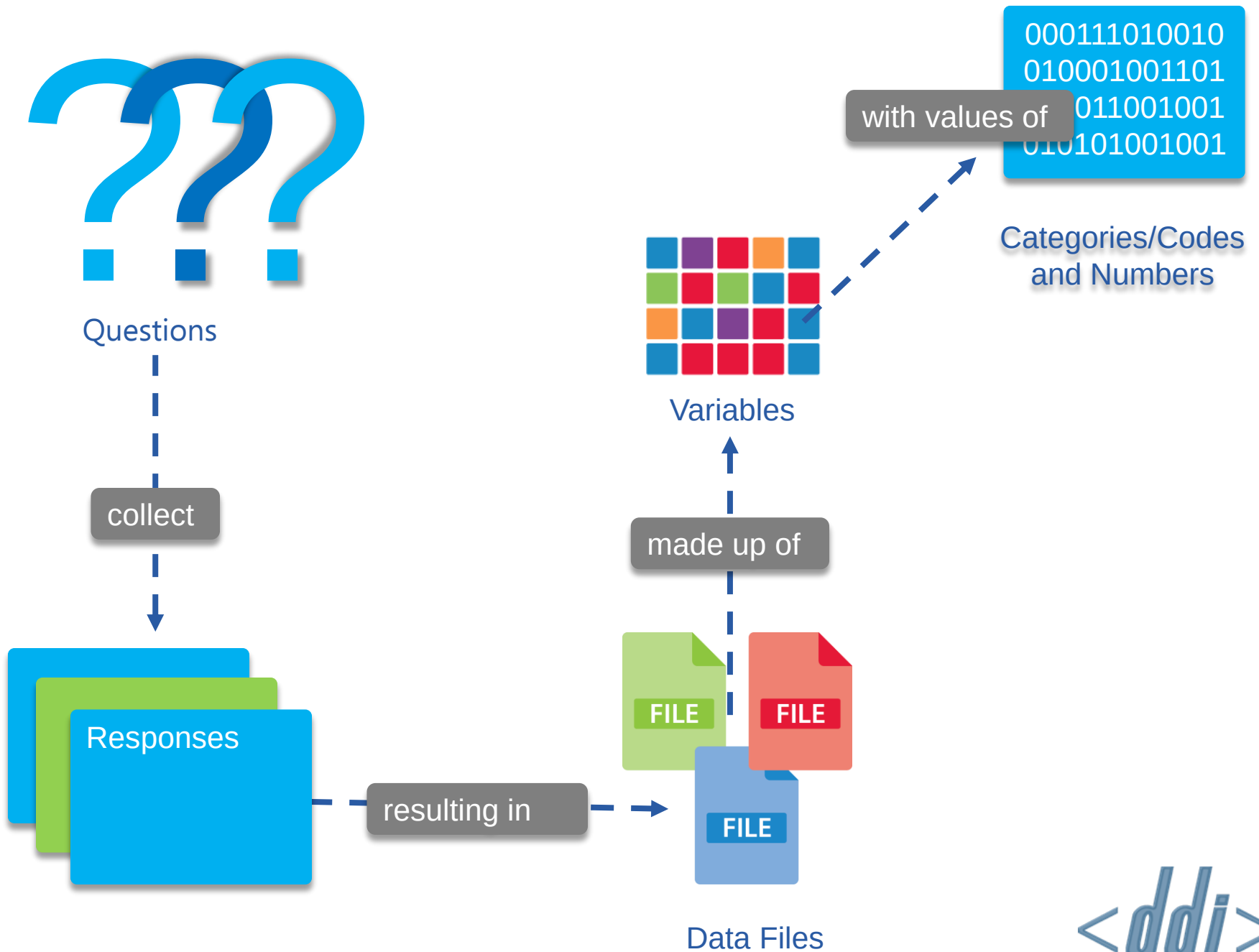
Introduction to Metadata

1. Usefulness of standardized metadata
2. Non-technical introduction to DDI
3. Opportunities created using standardized metadata
4. Case Studies



What do we mean by “metadata”?





DDI provides a vocabulary for this

What do we mean by “metadata”

- Information in three main areas
 - ▣ Survey design
 - ▣ Data documentation
 - ▣ Data lifecycle documentation

Data Documentation Initiative <ddi>

- Open standard for describing data
 - ▣ Focus on social, behavioral, and economic sciences
 - ▣ XML
- Users
 - ▣ National Statistical Institutes
 - ▣ University Research Groups
 - ▣ Data Archives
 - ▣ Other Data Producers and Publishers
- Since 1995

DDI Content

Study



Group



StudyUnit



Quality

Survey



DataCollection



Instrument



ControlConstructs



Question

Data



PhysicalInstance



DataRelationship



Variable

Foundational



Concept



Universe



Organization



CodeList



CategoryList



Category

Why Metadata?



- So researchers can understand the data and its context
- Adds credibility to the data

The Benefits of a Data Documentation Standard



- ❑ Share tools
- ❑ Share funding
- ❑ Benefit from other organizations' investments

Harmonization and Reuse

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Reuse any metadata element

- Question banks
- Variables banks
- Classification banks

Discover relationships among items

- Which datasets are based on the same questions?
- Which datasets have common variables
- Find any other relationships among items

Metadata-Driven Processes

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- DRY: Don't Repeat Yourself
 - ▣ Define things once and create multiple outputs from that canonical information
- Generate documentation as a byproduct of the process
- Populate CAI systems
 - ▣ Out of the box: Blaise, CASES, CPro, RedCAP, queXML
 - ▣ Custom systems: possible with addins

Some Use Cases

- ❑ Statistics New Zealand
- ❑ Statistics Denmark
- ❑ MIDUS
- ❑ CLOSER





Real-world Examples



FIELDS

```
Name      "What is your name?": STRING[20]
Sex        "What is your sex?": (Male, Female)
Age        "What is your age (in years)?": 0..120
MarStat    "What is your marital status?":
  (NeverMar "Never married",
   Married  "Married",
   Divorced "Divorced",
   Widowed  "Widowed")
PaidJob    "Do you have a paid job?": (Yes, No)
KindWork   "What kind of work do you do?": STRING[40]
Distance   "What is the distance to your work (in km)?": 0..300
Travel     "How do you travel to work?": SET [3] OF
```

Questions

Categories

TextDomain

NumericDomain

- POPULATION ELECTORAL
- Coverage
- Population and population forecasts
- Population in Denmark
- Population forecasts
- Immigrants and their descendants
- Births and adoptions
- Deaths and life expectancy
- Households, families and children
- Marriages and divorces
- Migrations
- Names

POPULATION IN DENMARK

Variable

Key Figures

POPULATION AT THE FIRST DAY OF THE QUARTER

Time: 2013Q2 | Unit: Number

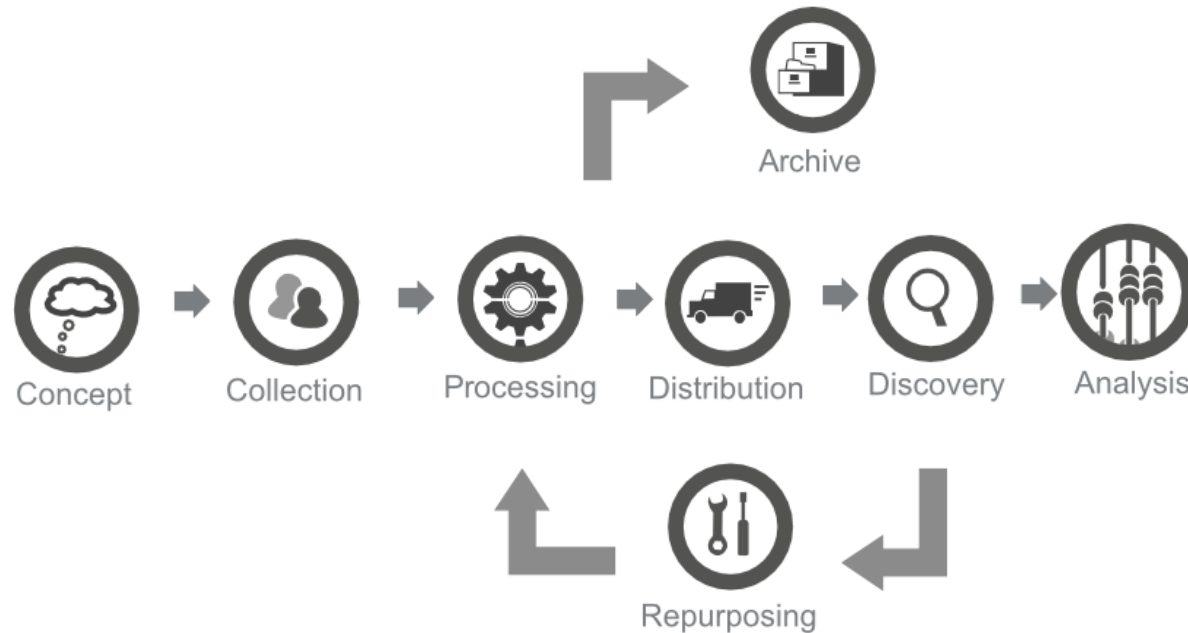
	Males	Females	Total
Total	2 180 576	2 825 260	5 605 836
0-9 years	328 422	312 500	640 922
10-19 years	354 242	337 050	691 292
20-29 years	348 907	338 939	687 846
30-39 years	348 614	347 209	695 823
40-49 years	411 408	403 416	814 824
50-59 years	365 489	363 646	729 135
60-69 years	342 568	352 122	694 690
70-79 years	195 802	223 620	419 422
80-89 years	74 692	116 781	191 473
90-99 years	10 275	29 163	39 438
100 years and more	157	814	971

Categories

SummaryStatistics

Why DDI?

- Document the full data lifecycle in a standard manner



Why DDI?

- Reusable metadata definitions
- No copy and paste
- Just point to an item
- Including metadata by reference helps avoid error and confusion
- Reuse is explicit

Metadata Banks



- DDI 3 supports the concept of metadata registries

Metadata Banks

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DDI 3 supports the concept of metadata registries

Reuse any metadata element

- Question banks
- Variables banks
- Classification banks

Discover relationships among items

- Which datasets are based on the same questions?
- Which datasets have common variables
- Find any other relationships among items

12/3/2015

Multilingual Support



- Most text fields can specify what language the content is in
 - These fields can be repeated to represent multiple languages
- 



Colectica Overview

Colectica Overview



- Standards-based metadata management
 - ▣ Survey design
 - ▣ Data documentation
 - ▣ Data lifecycle documentation

DDI Content

Study



Group



StudyUnit



Quality

Survey



DataCollection



Instrument



ControlConstructs



Question

Data



PhysicalInstance



DataRelationship



Variable

Foundational



Concept



Universe



Organization



CodeList



CategoryList



Category

The Colectica Platform

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Colectica Desktop Applications

- Create, ingest, manage, and edit metadata
- Usable DDI 3 for end-users

Colectica Repository

- Centralized, authoritative, metadata store built on DDI 3, ISO 11179, and Web Service standards

Colectica Portal

- Search and browse metadata from Colectica Repository

Colectica SDK

- Allows programmers to work with DDI 3 and interact with Colectica Repository

Colectica Toolkit

- Command line utilities to perform specific tasks

How to get information into DDI?

- Manually enter information
- Import (Colectica or Stat/Transfer)
 - Excel
 - Delimited Files
 - SPSS
 - Stata
 - Blaise
 - CASES
 - RedCAP
 - queXML
- Integrate custom data sources

Publish from Colectica

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Generate Paper Documentation

- Survey instruments, code lists, variable descriptions
- PDF, Word formats

Publish to Colectica Portal

- Full study descriptions
- Interactive browsing and discovery

Custom Publications

- Use Colectica Repository as a data source
- Custom web sites, paper formats, or other formats

Metadata-Driven Surveys

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- Populate CAI systems
 - ▣ Out of the box: Blaise, CASES, CSPro, RedCAP, queXML
 - ▣ Custom systems: possible with addins

Common Patterns in DDI

- Identification
- Naming
- Packaging

Common Elements

- Identification
 - ▣ Agency ID
 - ▣ Identifier
 - ▣ Version number

Common Elements: Descriptive

- Either
 - ▣ Name
 - ▣ Label
 - ▣ Description
- Or
 - ▣ Citation/Title (Dublin Core style)

DDI 3 Packaging

- **Fragment**
- OR: Modules + Schemes + Packages + Redundancy

In-depth: Data

Study



Group



StudyUnit



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StatisticalSummary

Foundational



Concept



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Organization



CodeList



CategoryList

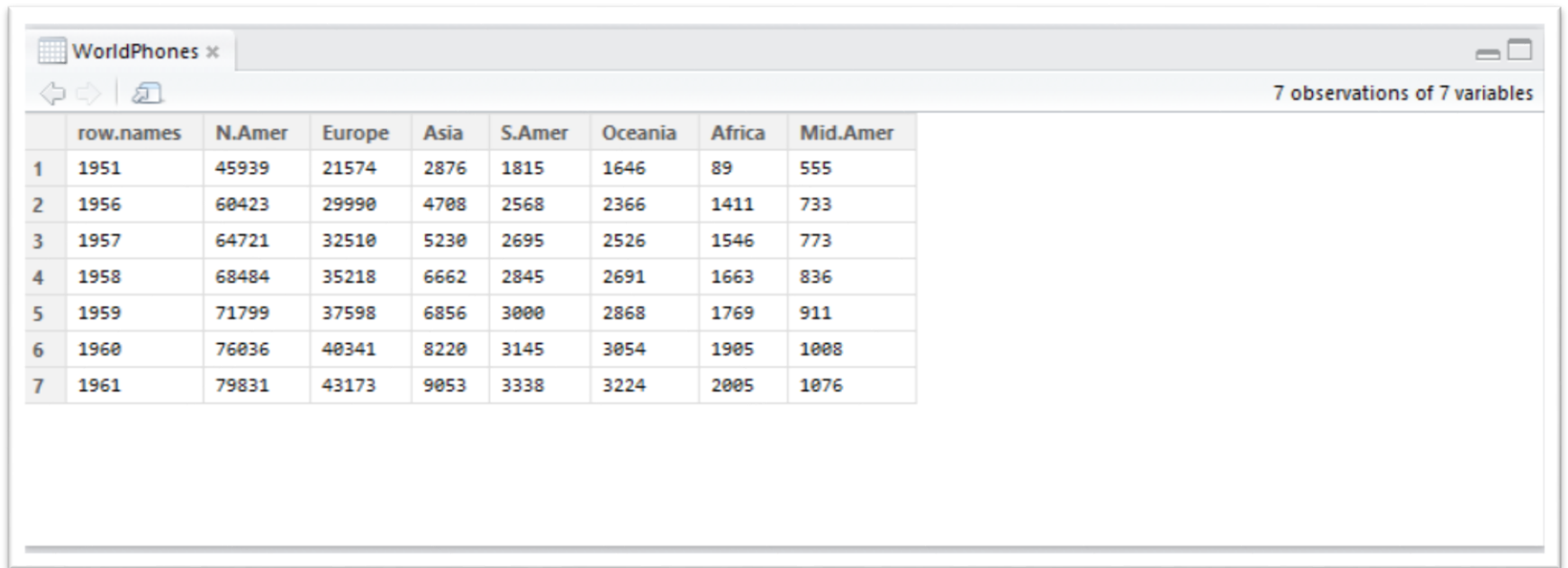


Category

Data

-  Dataset
 - A data file, database, or other source of data
-  Data Layout
 - Describes the layout of a data file
-  Variable
 - A column in a dataset
-  Variable Statistics
 - Summary statistics for a single variable
-  NCube
 - Aggregate data
-  Represented Variable
 - Describes the common information of one or more harmonized variables...
-  Conceptual Variable
 - Describes the common information of one or more harmonized variables

PhysicalInstance (Dataset)

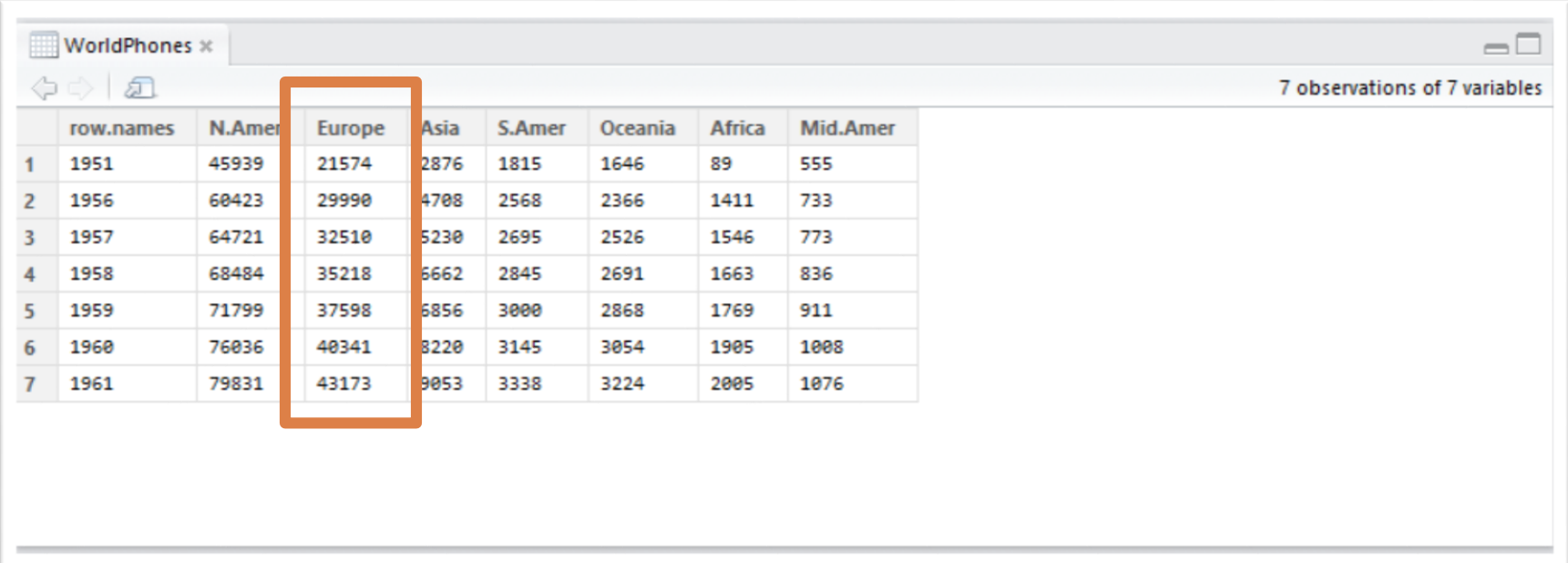


WorldPhones x

7 observations of 7 variables

	row.names	N.Amer	Europe	Asia	S.Amer	Oceania	Africa	Mid.Amer
1	1951	45939	21574	2876	1815	1646	89	555
2	1956	60423	29990	4708	2568	2366	1411	733
3	1957	64721	32510	5230	2695	2526	1546	773
4	1958	68484	35218	6662	2845	2691	1663	836
5	1959	71799	37598	6856	3000	2868	1769	911
6	1960	76036	40341	8220	3145	3054	1905	1008
7	1961	79831	43173	9053	3338	3224	2005	1076

Variable



WorldPhones x

7 observations of 7 variables

	row.names	N.Amer	Europe	Asia	S.Amer	Oceania	Africa	Mid.Amer
1	1951	45939	21574	2876	1815	1646	89	555
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6	1960	76036	40341	8220	3145	3054	1905	1008
7	1961	79831	43173	9053	3338	3224	2005	1076

StatisticalSummary

WorldPhones x

7 observations of 7 variables

	row.names	N.Amer	Europe	Asia	S.Amer	Oceania	Africa	Mid.Amer
1	1951	45939	21574	2876	1815	1646	89	555
2	1956	60423	29990	4708	2568	2366	1411	733
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6	1960	76036	40341	8220	3145	3054	1905	1008
7	1961	79831	43173	9053	3338	3224	2005	1076

Console ~/

```
> summary(worldPhones[3])  
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   
 64720  64720   64720   64720  64720   64720   
> |
```

Hands-on: Document a Dataset



- Have your own?
- Or, try:



Finding the Details



Finding the Details

- Sources
 - ▣ DDI Documentation
 - ▣ Colectica (or other tools)
 - ▣ XML Schemas

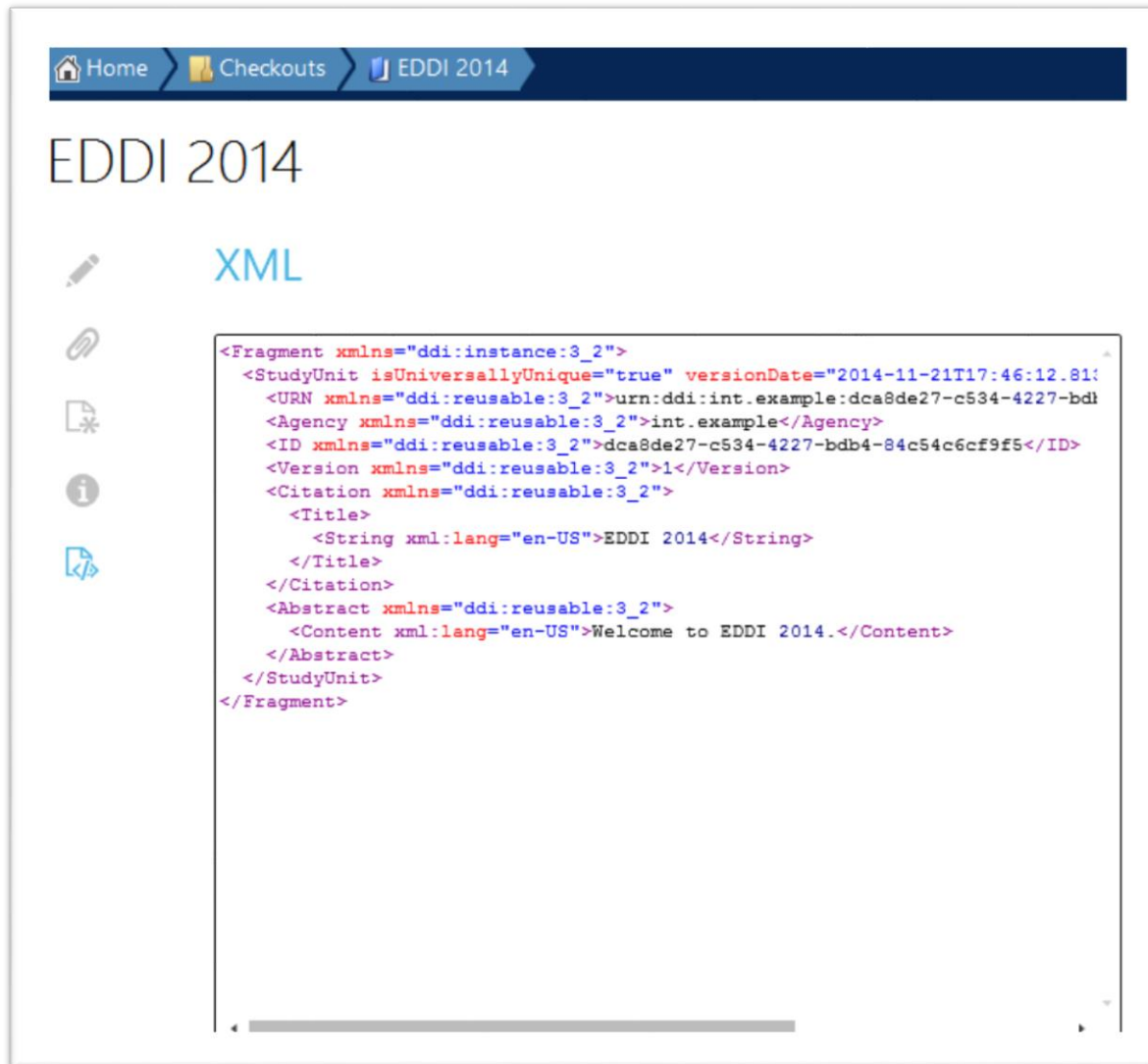
DDI Documentation



Available at <http://www.ddialliance.org/>



DDI in Colectica Reader and Designer



The screenshot shows the EDDI 2014 XML editor interface. At the top, a navigation bar contains links for Home, Checkouts, and EDDI 2014. Below the navigation bar, the title "EDDI 2014" is displayed. To the left of the XML editor is a sidebar with icons for editing (pencil), saving (floppy disk), deleting (trash), and viewing (eye). The main area displays the XML code for the EDDI 2014 instance.

```
<Fragment xmlns="ddi:instance:3_2">
  <StudyUnit isUniversallyUnique="true" versionDate="2014-11-21T17:46:12.814">
    <URN xmlns="ddi:reusable:3_2">urn:ddi:int.example:dca8de27-c534-4227-bd1
    <Agency xmlns="ddi:reusable:3_2">int.example</Agency>
    <ID xmlns="ddi:reusable:3_2">dca8de27-c534-4227-bdb4-84c54c6cf9f5</ID>
    <Version xmlns="ddi:reusable:3_2">1</Version>
    <Citation xmlns="ddi:reusable:3_2">
      <Title>
        <String xml:lang="en-US">EDDI 2014</String>
      </Title>
    </Citation>
    <Abstract xmlns="ddi:reusable:3_2">
      <Content xml:lang="en-US">Welcome to EDDI 2014.</Content>
    </Abstract>
  </StudyUnit>
</Fragment>
```

In-depth: Data

Study



Group



StudyUnit



Quality

Survey



DataCollection



Instrument



ControlConstructs



Question

Data



PhysicalInstance



DataRelationship



Variable



StatisticalSummary

Foundational



Concept



Universe



Organization



CodeList



CategoryList



Category

Foundational Metadata

- 💡 Concept
 - An abstract idea or general notion
- 📋 Category
 - A class of people or things
- 📄 Code List
 - A list of categories, each with an assigned value
- 🌐 Universe
 - A population being studied
- 🏢 Organization
 - An institution, company, or other group



Classifications in DDI



Classifications in DDI

- DDI 2 and DDI 3 do not directly support this
- But other things do:
 - ▣ Neuchâtel
 - ▣ GSIM
- DDI 3 is quite extensible

Copenhagen Mapping

- Map the Neuchâtel-GSIM classifications model to DDI 3.2
- Built around:
 - ▣ CodeLists
 - ▣ Categories
 - ▣ Groups of items with well-defined terms from controlled vocabularies
- DDI 3.3 should support this model without extensions

Metadata Publication

PDF

Repository

Web

Publish to PDF



Publish to a Repository



Publish to the Web



Hands-on: Publish

- Publish to PDF
- Publish to the Web

Repository Information



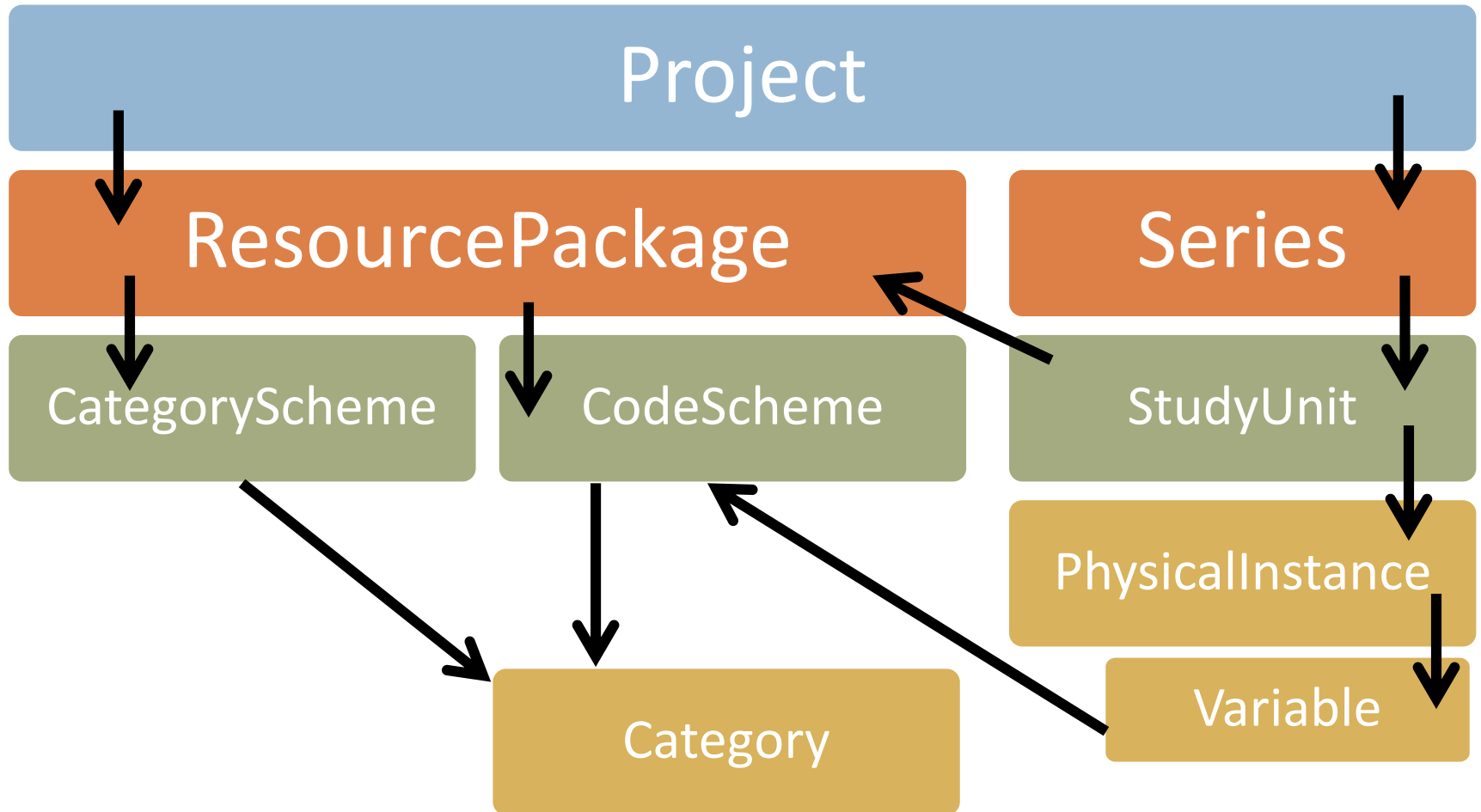
- ❑ Hostname: `workshop.colectica.org`
- ❑ User Name: `workshop`
- ❑ Password: `iassist`



DDI Versioning



Version Propagation



Colectica takes care of
this for you

In-depth: Study Lifecycle

Study



Group



StudyUnit



Quality

Survey



DataCollection



Instrument



ControlConstruct
s



Question

Data



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CodeList



CategoryList



Category

Study

Colectica lets you document both individual and repeated studies.

- Study

- A single research project

- Series

- A repeated set of studies

- Archive

- Information about how a study is archived for long term preservation

StudyUnit

Yale University

Calendar | A-Z Index

Yale

ISPS

Institution for Social and Policy Studies

ADVANCING RESEARCH • SHAPING POLICY • DEVELOPING LEADERS

HOME

RESEARCH

NEWS

EVENTS

PROGRAMS

TEAM

ABOUT

Areas of Study

Field Experiments Initiative

Publications

Data

Ongoing Projects & Studies

Science Deserves Better: The Imperative to Share Complete Replication Files

AUTHOR(S):

?

Allan Dafoe

ISPS ID:

?

D108

RELATED PUBLICATIONS:

Science Deserves Better: The Imperative to Share Complete Replication Files

KEYWORD(S):

?

Replication

RESEARCH DESIGN:

?

Observational

DATA TYPE:

Survey; Administrative

DATA SOURCE(S):

?

Author.

DATA SOURCE INFORMATION:

?

Author.

(1) Data is of scholars responding to a survey about their experiences trying to replicate published quantitative work.

(2) Observations are articles published in APSR or AJPS in recent years. Variables code whether replication files are available.

Suggested citation: "Dafoe, Allan (2014). Replication Materials for: 'Science Deserves Better: The Imperative to Share Complete Replication Files,' <http://hdl.handle.net/10079/66t1gdc>. ISPS Data Archive."

FIELD DATE:

?

December 1, 2013

LOCATION:

United States

SEARCH ISPS

ISPS KNOWLEDGEBASE

Yale ISPS KnowledgeBase

Data

Projects

Publications

Terms of use

About the ISPS Data Archive

Author

Any

Project Area

Any

Keywords

Any

Year

Any

Keywords

Any

Keywords

Any

Search

See all data

The ISPS KnowledgeBase is the gateway to all ISPS data, projects, and publications. It is an integrated database which provides a one-stop-shop for ISPS-related research products.

Search the KnowledgeBase or browse recent additions.

StudyUnit

- EDDI 2015
 - ▣ A study of EDDI workshop attendees in Copenhagen

Universe (Population)

DATA SOURCE INFORMATION:	① Author. (1) Data is of scholars responding to a survey about their experiences trying to replicate published quantitative work. (2) Observations are articles published in APSR or AJPS in recent years. Variables code whether replication files are available. Suggested citation: "Dafoe, Allan (2014). Replication Materials for: 'Science Deserves Better: The Imperative to Share Complete Replication Files,' http://hdl.handle.net/10079/66t1gdc. ISPS Data Archive."
FIELD DATE:	① December 1, 2013
LOCATION:	United States
UNIT OF OBSERVATION:	① (1) scholars who attempted replications, (2) published articles
SAMPLE SIZE:	① (1) 190, (2) 342
INCLUSION/EXCLUSION:	① (1) Three groups of scholars were surveyed about their experiences attempting to replicate statistical studies: students from the author's PhD methods class, students from Gary King's PhD methods class, and subscribers to the Political Methodology listserve, (2) Data was collected on the availability of replication files for recent publications in the two top political science journals, the American Political Science Review (APSR) since 2010 and the American Journal of Political Science (AJPS) since 2009.

Universe



Universe (Population)

- People in this room, right now

Group (Series)



The screenshot displays the DataInfo+ interface for the Consumers Price Index. The header includes the Statistics New Zealand logo and the DataInfo+ title with the tagline 'Explore our metadata'. Navigation links for Home, Glossary, User guide, and Search are present. A left sidebar lists categories: Consumers Price Index, Classifications (0), Concept Sets (1), Data Collections (1), Population Sets (1), Variable Groups (0), and Variable Sets (1). The main content area for the Consumers Price Index includes an Abstract, Purpose, Citation Information, Coverage Information, and Temporal/Topical Coverage details.

Statistics New Zealand
TAHIRANGA Kaitiaki

DataInfo+
Explore our metadata

Home Glossary User guide Search

Consumers Price Index

Abstract

The CPI measures the rate of price change of goods and services purchased by New Zealand households.

Purpose

The consumers price index (CPI) measures the changing price of a fixed basket of goods and services purchased by New Zealand households. The aim of the CPI is to measure price changes of the same sample of products at each outlet over time. When there is a change in the size or quality of any of the goods or services in the basket, an adjustment is made to ensure that the price change shown in the CPI is not affected by the change in size or quality.

The CPI represents \$88.9 billion spent on goods and services by New Zealand households, at June 2011 quarter prices. This is based on information from the 2009/10 Household Economic Survey and other sources.

The CPI is used to help set monetary policy and for monitoring economic performance. It is used by the government to adjust New Zealand Superannuation and unemployment benefit payments once a year, to help ensure that these payments maintain their purchasing power. Employers and employees use the CPI in wage negotiations.

The CPI is published quarterly. The food group is the only CPI group for which an index is published each month. This is known as the Food Price Index.

Citation Information

Title
Consumers Price Index

Alternate Title
CPI

Creator
Prices

Publisher
Statistics New Zealand

Rights
Statistics New Zealand

Coverage Information

Temporal Coverage

- 1914 to

Topical Coverage

- Price indexes
- Consumers Price Index (CPI)
- Food Price Index (FPI)

Group (Series)

- European DDI User Conference
 - ▣ IASSIST 2012
 - ▣ IASSIST 2013
 - ▣ IASSIST 2014
 - ▣ IASSIST 2015
 - ▣ IASSIST 2016

Hands-on: Document a Study

- Have your own?
- Or, try:

In-depth: Surveys

Study



Group



StudyUnit



Quality

Survey



DataCollection



Instrument



ControlConstructs



Question

Data



PhysicalInstance



DataRelationship



Variable

Foundational



Concept



Universe



Organization



CodeList



CategoryList



Category

Data Collection



Instrument

A survey or other data capture instrument



Question

A question that can appear in a survey instrument



Question Grid

A question grid that can appear in a survey instrument



Question Block

A question block that can appear in a survey instrument



Statement

A statement that can appear in a survey instrument



Instruction

Information for an interviewer or respondent



Computation

Source code that performs calculations, validation, or other actions



Sequence

A set of items in an instrument, used for grouping, paging, or other organization



Data Collection

Describes the processes and methods used to collect data

Questions

- What is your name?
- How did you get here?

A Question in DDI

```
<d:QuestionItem>
```

```
  <r:Agency>example.org</r:Agency>
```

```
  <r:ID>q1</r:ID>
```

```
  <r:Version>1</r:Version>
```

```
  <d:QuestionItemName xml:lang="en">
```

```
    <r:String>name</r:String>
```

```
  </d:QuestionItemName>
```

```
  <d:QuestionText audienceLanguage="en">
```

```
    <d:LiteralText>
```

```
      <d:Text>What is your name?</d:Text>
```

```
    </d:LiteralText>
```

```
  </d:QuestionText>
```

```
  <d:TextDomain minLength="1"></d:TextDomain>
```

```
</d:QuestionItem>
```

A Question in DDI

```
<d:QuestionItem>
```

```
  <r:Agency>example.org</r:Agency>
```

```
  <r:ID>q1</r:ID>
```

```
  <r:Version>1</r:Version>
```

```
  <d:QuestionItemName xml:lang="en">
```

```
    <r:String>name</r:String>
```

```
  </d:QuestionItemName>
```

```
  <d:QuestionText audienceLanguage="en">
```

```
    <d:LiteralText>
```

```
      <d:Text>What is your name?</d:Text>
```

```
    </d:LiteralText>
```

```
  </d:QuestionText>
```

```
  <d:TextDomain minLength="1"></d:TextDomain>
```

```
</d:QuestionItem>
```

Other Response Types

- Text
- Numeric
- DateTime
- Code

Question

- Does your organization currently use DDI for any purpose?
 - ▣ Yes
 - ▣ No

Instrument

B19.

[B1PB19] Are you married, separated, divorced, widowed, or never married?

1. MARRIED
2. SEPARATED
3. DIVORCED
4. WIDOWED
5. NEVER MARRIED
7. DON'T KNOW/NOT SURE
8. REFUSED
9. INAPP

B20.

[B1PB20] How many times have you been married altogether?

- # TIMES MARRIED
97. DON'T KNOW/NOT SURE
 98. REFUSED
 99. INAPP

B21MO.

[B1PB21M] In what month and year were you married (for the first time)?

(MONTH)

- | | |
|--------------|-------------------------|
| 1. JANUARY | 10. OCTOBER |
| 2. FEBRUARY | 11. NOVEMBER |
| 3. MARCH | 12. DECEMBER |
| 4. APRIL | 97. DON'T KNOW/NOT SURE |
| 5. MAY | 97. REFUSED |
| 6. JUNE | 98. INAPP |
| 7. JULY | |
| 8. AUGUST | |
| 9. SEPTEMBER | |

Instrument



- Let's make a quick survey for the workshop

ControlConstructs

- If the organization uses DDI, ask:
 - ▣ Which parts?
 - Survey
 - Data
 - Study Lifecycle
 - Other

ControlConstructs

The screenshot displays a form builder interface with three main sections: 'Age', 'Action', and two 'Conditional' blocks. The 'Age' section contains a question 'What is your age?' with a 'Numeric' input field. The 'Action' section is highlighted in blue and contains a 'Conditional' block with two items: 'Age >= 14' (3 items) and '((Age <> EMPTY) AND (Age < 14))' (1 item). The first 'Conditional' block is highlighted in blue, and the second 'Conditional' block is below it. Arrows point from labels to specific elements: 'Question' points to the 'What is your age?' text, 'IfThen' points to the 'Age >= 14' condition, and 'Sequence' points to the '3 items' text under the first condition.

Age

What is your age?

Numeric

Action

Conditional

- Age >= 14
3 items
- ((Age <> EMPTY) AND (Age < 14))
1 items

Conditional

- Job = Yes
3 items
- ((Job <> EMPTY) AND (Job = No))
1 items

Demonstration



Hands-on: Make a Survey



- Have your own?
- Or, try:



Longitudinal Data

Another Quick Survey



<http://bit.ly/eddi2015-workshop-2>

DDI Model for Repeated Data

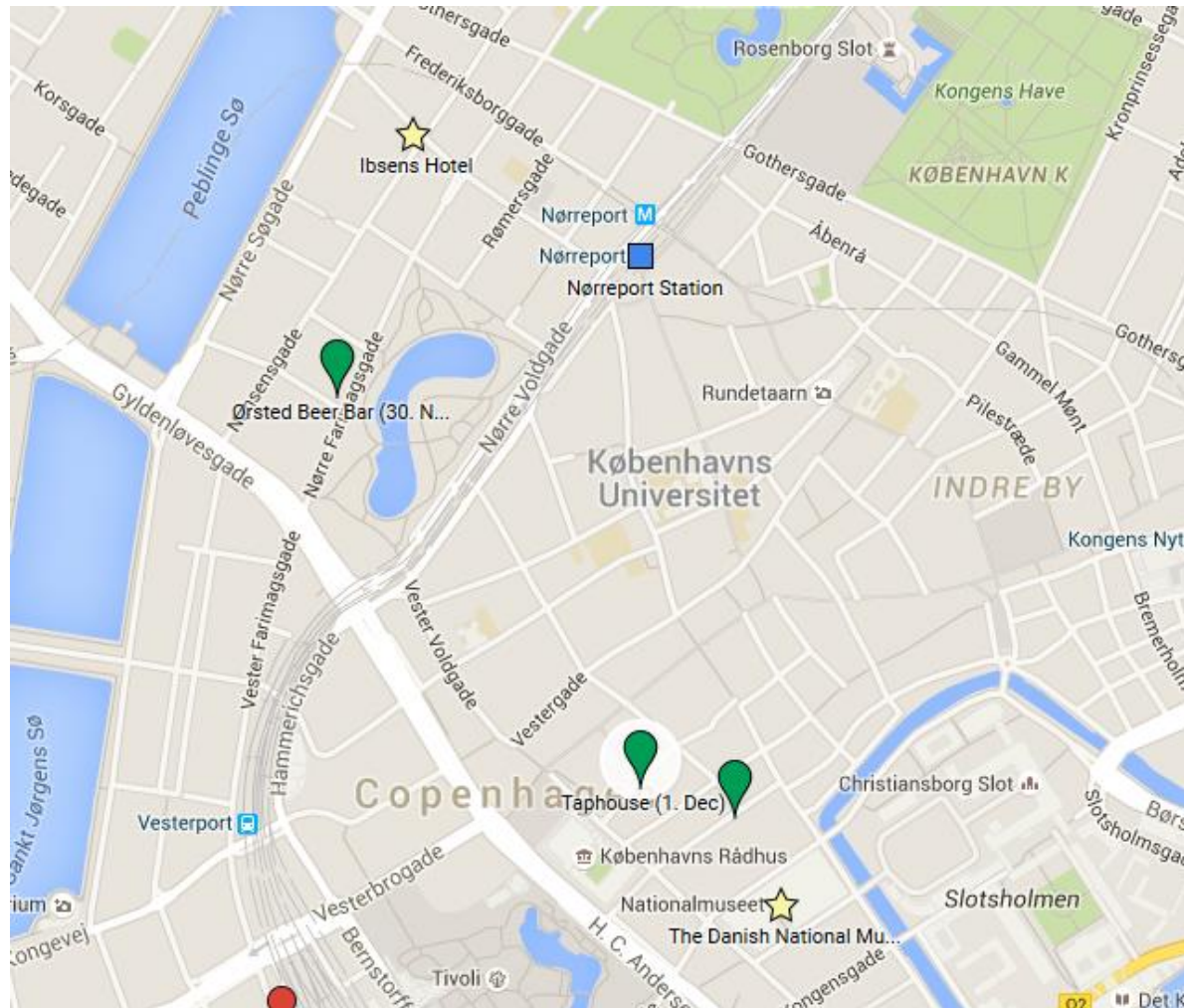
- Variable
- RepresentedVariable
- ConceptualVariable

Recap

- Colectica and DDI let you document:
 - ▣ Data
 - ▣ Surveys
 - ▣ Study Lifecycle
- Publish to PDF and the Web

“Informal Gathering”

Location: Taphouse



Feedback

bit.ly/eddi2015-workshop-feedback





Photo by johannes on flickr

Thank You

Jeremy Iverson

jeremy@colectica.com

Dan Smith

dan@colectica.com